Shashank Singh Shekhawat

Technical Lead – Hardware Electronics & Battery Management

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Professional Profile

Hardware electronics engineer and Technical Lead with more than 7 years of engineering experience in automotive electronics, battery management systems (BMS) and power electronics. Develops low-voltage (up to 60 V) and high-voltage (400/800 V) hardware for automotive ECUs, from LV sensing and control electronics to safety-relevant HV BMS hardware. Technical interface between systems, software, safety, testing, manufacturing and suppliers.

Results at BMW: Redesigned the Gen6 hardware architecture (up to 15% lower BOM and PCB-development cost); led hardware development of a safety-relevant BMS module from B-sample to Start of Production (SOP) with zero safety escapes, coordinating 7 engineers; eliminated a systematic offset error in battery-current measurement.

Professional Experience

BMW Group, Munich, Germany **Apr. 2023 – Present**

Technical Lead – Electronics Hardware Development

- Directed hardware concepts for safety-relevant control and sensing electronics on the HV BMS platform (ASIL C).
- Owned the full hardware lifecycle: requirements analysis, circuit design, component selection, schematic capture, design freezes, PCB layout governance, manufacturing-data release, PCBA assembly and verification for series production.
- Redesigned the Gen6 ECU hardware architecture, reducing component count and BOM/PCB-development cost by up to 15% while improving thermal performance, reliability and robustness.
- Coordinated a cross-functional team of 7 engineers across systems, embedded software, mechanics, functional safety, quality, testing and manufacturing; achieved SOP of the module with zero safety escapes.
- Improved battery-current measurement accuracy by eliminating a systematic offset error in the sensing chain, improving vehicle-range prediction and closing a critical product issue for the next-generation BEV platform.
- Led component engineering for key power and sensing devices: datasheet evaluation, SOA/derating review, worst-case tolerance analysis and Monte-Carlo simulation, second-source qualification and PCN/obsolescence management with semiconductor suppliers.
- Authored technical specifications and RFQs, evaluated supplier offers and competing technical solutions, steered Tier-1, EMS and PCB suppliers, contributed to technology roadmaps and presented business cases to senior management.
- Designed H-bridge DC motor-control circuitry for the HV BMS and worked with Gen6 cell-monitor ICs.
- Supported ISO 26262 (ASIL C) via FMEDA, single-point-fault analysis and safety reviews; supported EMC robustness via filter design, PCB reviews and environmental testing (temperature cycling, vibration, transients, overvoltage, short-circuit, reverse polarity).

BMW Group, Munich, Germany **Jun. 2022 – Mar. 2023**

Master Thesis – Power Electronics

- *Development and Simulation of Electronic Fuses in LV Automotive Battery Systems*: modelled eFuse switching dynamics and fault response to support automotive power-protection design decisions.
- Built a MATLAB/Simulink simulation framework for eFuse behaviour under load and fault conditions, enabling repeatable evaluation of switching stress, transient behaviour and device limits.

Danfoss Silicon Power, Munich, Germany **Dec. 2020 – May 2022**

Working Student – Power Electronics

- Modelled and simulated power-converter topologies for electric-vehicle and trolley-system applications in PLECS and MATLAB/Simulink.
- Developed onboard-charging concepts and parameter calculations; implemented control and modulation strategies for high-power rectifiers and inverters.
- Supported power-module testing on an automotive double-pulse test bench, including measurement setup, data evaluation and documentation.
- Designed an optical-fibre converter board for a SiC gate-driver circuit and assisted with analog, digital and power-electronics assembly during lab validation.

Würth Elektronik, Munich, Germany **Apr. 2021 – Sep. 2021**
Intern – Power Management and Energy Harvesting

- Developed a power-management circuit for an energy-harvesting device for sensor and IoT applications.
- Designed analog MPPT circuitry for DC-DC converters in Altium Designer and simulated performance in LTspice.
- Performed line- and load-regulation testing on power-module evaluation boards with spectrum analysers, electronic loads and standard lab instrumentation.

Stabl Energy, Munich, Germany **May 2020 – Oct. 2020**
Working Student – Power Electronics Development

- Designed and modelled inverter concepts for battery-storage modules and developed modulation approaches for power-converter applications.
- Built, soldered, tested and debugged power-electronics prototypes with oscilloscopes, LCR meters and spectrum analysers; documented measured data for design improvements.
- Designed and built a high-voltage Line Impedance Stabilisation Network (LISN) for EMC pre-compliance evaluation of inverter systems.

Bosch Engineering and Business Solutions, Bangalore, India **Aug. 2017 – Jul. 2019**
Associate Software Engineer – Automotive Systems

- Supported functional development of ADAS and battery-management features for a premium automotive OEM's hybrid and electric vehicles.
- Contributed to schematic-related activities, software integration and ECU simulation for hybrid and electric drivetrains, including calibration and testing on vehicle simulators and target vehicles.
- Performed Hardware-in-the-Loop (HIL) functional testing with CANoe and ECU-TEST; automated and benchmarked test activities with scripts in a real-time operating-system environment.

Education

Technical University of Munich (TUM), Germany **Oct. 2019 – Mar. 2023**
M.Sc. in Power Engineering *Grade: 1.8*
Focus areas: High-Voltage Engineering, Energy Storage Systems and Smart Grids. **Deutschlandstipendium** (2021 – 2022).

Strascheg Center for Entrepreneurship (HM), Germany **Oct. 2021 – Aug. 2022**
Academic Program for Entrepreneurship (APE)
Focus areas: Design Thinking, Lean Start-Up, Cross-Functional Leadership, Business Modelling and Rapid Prototyping.

Manipal Institute of Technology, India **Aug. 2013 – Jun. 2017**
B.E. in Electrical and Electronics Engineering *Grade: 1.7*
Focus areas: Network Analysis, MEMS Technology, Embedded Systems and Control Devices.

Skills & Languages

Hardware & Domains	:	Power electronics, battery management systems (LV & 400/800 V HV), automotive ECUs, analog & mixed-signal circuits, sensing & measurement electronics, EMC, ISO 26262 Functional Safety (IEC 61508), AEC-Q100/Q101 components.
Tools & Technologies	:	Altium Designer, LTspice, MATLAB/Simulink, PLECS, SIMetrix, Eagle, CANoe, ECU-TEST, Jira.
Communication Protocols	:	CAN, LIN, SENT, SPI, FlexRay.
Lab & Validation	:	Oscilloscopes, power and spectrum analysers, LCR meters, electronic loads, soldering & debugging.
Programming	:	C/C++, Python, VBA, embedded programming.
Languages	:	English (C2), German (B2), Hindi (native).

Engagement & Interests

Engagement: Resident Tutor (Munich Student Union; 2020–2022); Core Committee Head (IEEE Student Branch; 2015–2016). **Interests:** Badminton, cycling, hiking, travelling.